



HEAT-TREAT REFERENCE

22 steels. Working recipes, plain-words theory, quench guide, troubleshooting, and the habits that keep blades out of the scrap bucket. Temperatures in °F. Temper twice, two hours per cycle, cool to room temperature between cycles.

WHAT'S INSIDE

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HOW TO USE THIS BOOKLET

1	Find your steel on page 3 or 4. Run the recipe as written on a TEST COUPON first (page 5).
2	File test + snap test the coupon. Happy? Commit the blade. Not happy? Adjust $\pm 25^{\circ}\text{F}$ and re-test.
3	Log every run — steel, batch, numbers, result. The knife that came out perfect is only repeatable if you wrote it down.

Read this once, believe it forever: these are proven STARTING points, not gospel. Ovens lie by $\pm 25^{\circ}\text{F}$, steel batches vary, and blade geometry changes everything. Your steel supplier's datasheet outranks this booklet; your own test coupon outranks both.

Oven vs forge: everything on page 4 (stainless/PM) needs a real oven with soak control and foil. Page 3 steels can be done by eye in a forge — non-magnetic plus a shade — but an oven turns "usually fine" into "every time."

THE FIVE STEPS, IN PLAIN WORDS

STEP	WHAT ACTUALLY HAPPENS	WHAT IT DOES FOR THE BLADE
Normalize	Heat past critical, air cool. Repeat 2–3× stepping down.	Resets grain after forging. Fine grain = tough blade. Skip it on forged blades and the quench finds out.
Austenitize	Hold at the recipe temperature so carbon dissolves into the iron.	Loads the spring. Too cool = carbon stays parked = soft. Too hot = grain grows = brittle.
Quench	Cool faster than the steel can relax — oil, plates, or air by steel.	Traps the carbon: martensite. This is the hardening moment. Speed must MATCH the steel.
Temper	Bake at 300–500°F, twice, 2 h each.	Trades a little hardness for a lot of toughness. An untempered blade is glass with a handle.
Cryo / sub-zero	Dry ice or liquid nitrogen between quench and temper.	Stainless/PM steels keep some soft “retained” structure after the quench — cold finishes the job. +0.5–1.5 HRC.

QUENCH MEDIA — MATCH THE SPEED TO THE STEEL

MEDIUM	SPEED	FOR	WATCH OUT
Brine / water	Violent	W2 hamons, W1	Highest crack risk. Know why you’re using it.
Fast oil (Parks 50 class)	Fast	1084 · 1095 · 15N20 · 26C3 · W2 · 52100	Shallow-hardening steels NEED this — canola gives 1095 soft spots.
Medium oil (AAA class · warm canola)	Medium	O1 · 5160 · 80CrV2 · 8670 · 1075	Warm the oil to ~120°F. Cold thick oil = slow = soft.
Aluminum plates + air	Steel-dependent	Every steel on page 4, plus A2 · D2	Clamp fast — thin blades warp less between plates than in oil.

TEMPER TEMPERATURE → HARDNESS (CARBON STEELS, ROUGH MAP)

TEMPER 2×2H	300°F	350°F	400°F	450°F	500°F
ROUGH HRC	63–64	61–62	60–61	58–59	56–57
FEELS LIKE	Razor, chippy	Fine edge	The sweet spot	Working tough	Chopper tough

Stainless/PM steels don’t follow this map — most run 300–350°F for kitchen hardness, and the tool-steel PMs (CruWear, 3V) like 400–500°F. Use the per-steel numbers on pages 3–4.

CARBON & LOW-ALLOY STEELS — °F

STEEL	NORMALIZE	AUSTENITIZE · SOAK	QUENCH	TEMPER	HRC	NOTES
1075	1600	1475–1500 · 10 min	Fast-med oil	375–425	56–58	Big blades, choppers, forgiving as it gets.
1084	1600	1475–1500 · 10–15	Fast oil	375–425	58–60	The beginner benchmark. Clean and honest.
1095	1650	1475 · 10	FAST oil only	350–400	60–62	Slow oil = soft spots. Thin edges warp — straighten between tempers.
15N20	1600	1475–1500 · 10	Fast oil	350–400	58–61	Nickel-tough. The bright layer in damascus.
26C3	1575	1450–1475 · short	Fast oil / brine	300–350	62–64	Razor carbon. Decarbs fast — coat or foil.
80CrV2	1600	1500–1525 · 10–15	Fast-med oil	400–450	57–59	Tough field-knife favorite.
8670	1600	1475–1525 · 10	Medium oil	350–425	58–60	Band-saw steel. Absurdly tough, cheap.
5160	1650	1525 · 10–15	Medium oil	400–450	57–58	Spring steel — swords and choppers.
52100	1650 / 1600 / 1550	1500–1550 · short	Fast oil	350–400	60–62	Triple normalize pays. Fine carbide, fine edge.
W2	1600	1450–1475 · 10	Brine / fast oil	350–400	60–62	Shallow hardening on purpose — the hamon steel.
O1	1600	1450–1500 · 10–15	Medium oil	~400	60–61	Forgiving tool steel for a basic shop.
A2	—	1750–1775 · 20–30 · foil	Plate / air	400–450	59–61	Air hardens. Stable in the quench, good for thin grinds.
D2	—	1850 · 30–45 · foil	Plate / air	400–500	58–61	Semi-stainless, carbide-rich, toothy working edge.

Soak time means AT temperature, not “in the oven.” The blade needs 2–4 minutes to catch up every time the door opens. Thin coupon ≠ thick chopper — add soak for mass.

STAINLESS & POWDER STEELS — °F · ALL FOIL-WRAPPED · ALL PLATE-QUENCHED

STEEL	AUSTENITIZE · SOAK	CRYO	TEMPER	HRC	NOTES
AEB-L	1950 · 10-15	Recommended	300-350	60-62	Stainless that behaves like carbon. Kitchen royalty.
14C28N	1950 · 10-15	Recommended	300-350	60-62	Budget stainless that punches way up.
Nitro-V	1950 · 10-15	Recommended	300-350	60-62	AEB-L's nitrogen cousin — a touch more rust resistance.
CPM-154	1950-1975 · 30	Recommended	400-500	58-61	The gentleman's stainless. Mirror polishes.
S35VN	1950-2000 · 30	Yes	300-350	59-61	Production-knife standard. Balanced everything.
Elmax	1950-1975 · 30	Yes	300-350	59-61	Euro powder stainless — wear + rust resistance.
MagnaCut	1950-2050 · 15-30	Strongly rec.	300-350	60.5-63.5	The modern do-it-all. Salt-water safe.
CruWear	1950-2050 · 15-30	Recommended	400-500	60-62	Tool PM — holds a 16° hunter edge through bone.
CPM-3V	1875-1950 · 20-30	Recommended	400-500	58-60	The toughness king of the PMs. Hard-use fixed blades.

- Foil, always:

above -1800°F bare steel sheds carbon (decarb) and scales. Double-fold stainless foil; from foil to plates in seconds.
- Cryo, honestly:

dry ice + kerosene slurry (~-100°F) captures most of the benefit for a hobby shop; liquid nitrogen is the full send. Right after the plates, before the first temper.
- Two temper camps on this page:

kitchen crowd (300-350°F, max hardness) and tool crowd (400-500°F, max toughness — CruWear, 3V, CPM-154). Don't split the difference; pick the blade's job.

TROUBLESHOOTING — SYMPTOM → USUAL SUSPECT → FIX

SYMPTOM	USUAL SUSPECT	FIX
File bites after quench	Under-heated, short soak, or quench too slow for the steel	Re-normalize, re-run 25°F hotter or with faster oil. Check the oven against a known melt point.
Soft spots	Slow oil on a shallow-hardening steel (1095, W2, 26C3)	Fast oil, better agitation — figure-eights, never side to side.
Warp	Uneven entry, uneven grind, edge too thin before HT	Enter vertically, spine-first slightly. Leave the edge a dime thick. Straighten between tempers at ~400°F, gloved.
Crack — PING	Quench too violent, sharp plunge corners, overheated grain	Radius every inside corner before HT. The right oil, not the fastest oil. Normalize properly.
Grey crust / soft skin	Decarb — bare steel at high heat too long	Foil (stainless) or anti-scale coating (carbon). Grind past the skin before judging hardness.
Chippy edge in use	Tempered too low for the job, or coarse grain	Re-temper 25–50°F up — costs one HRC point, saves the edge. Coarse grain: start over, normalize 3x.
Won't hold an edge	Over-tempered, or the decarb skin IS the edge	Fresh coupon, file test at each temper step, find where it went soft.

THE TEST-COUPON PROTOCOL — 20 MINUTES THAT SAVES BLADES

1	Cut a coupon from the SAME bar, same thickness as your edge. Mark the batch on it.
2	Run the full recipe on the coupon — normalize included if you forged.
3	File test as-quenched: the file must skate. Snap it in a vise, read the grain: silky grey = fine; sparkly crystals = coarse, start over.
4	Temper a second coupon, file test again — it should barely bite. Now commit the blade.

HABITS THAT SAVE BLADES

Snap temper	Within the hour after quench: 300–350°F, 30 min — BEFORE straightening or grinding.
Warm the oil	Quench oils want ~100–130°F. Cold oil is slow oil.
Calibrate the oven	Table salt melts at 1474°F — a pinch on a coupon tells you if your dial lies.
Write it down	Steel, batch, numbers, result — every run. The Forge Logbook pairs with this booklet.